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Review

Numerical resolution of the liquid–vapour two-phase flow by means of the two-fluid model and a pressure based method

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ABSTRACT

A numerical study based on a two-fluid model to describe the fluid-dynamic behaviour of the two-phase flow inside ducts is presented. The discretization of the governing equations has been developed by means of the finite volume technique using a staggered mesh. A semi-implicit pressure-based method is used to couple the fluid conservation equations. Different empirical correlations have been used to evaluate the mass, momentum and energy exchanged through the interface, gas and liquid distribution in the tube, and other terms which appear in the conservation equations. These correlations are used depending on the flow regime map, which is function of the gas volume fraction and velocities. The instability of the two-fluid model due to the equations system is non-hyperbolic has been analyzed. Different two-phase flow cases have been solved by means of the method proposed in this paper. After a verification process to assess the quality of the numerical results obtained, the results are compared, when possible, with analytical solutions, and also with experimental data obtained by the authors.

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1. Introduction

In different industrial applications the two-phase flow phenomena is very common. It can be described as the flow of two different kind of matter, e.g. solid particles in a gas or liquid, gas bubbles in a liquid, or droplets in a gas stream. Two-phase flow is present in many manufacturing processes, and a variety of fluids can be working at the same time, e.g. air–water, oil–water, carbon particles-steam, etc. (Kleinstreuer, 2003). The high number of applications, where the two-phase flow phenomena is present, has increased the motivation to study the multiphase flow in multicomponent fluids during the last 40 years.

A deeper understanding of the two-phase flow phenomena (boiling or condensation) helps to find new solutions and improve the design of the refrigerant and air-conditioning system. Usually, this task uses a resolution of a mathematical model as a power tool

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Nomenclature

S	cross section area of the flux (m^2)
A_t	cross section area in the tube (m^2)
D	diameter (m)
CV	control volume
e	specific total energy ($J\,kg^{-1}$)
g	acceleration of gravity ($m\,s^{-2}$)
G	mass flow rate per area ($kg\,s^{-1}\,m^{-2}$)
h	specific enthalpy ($J\,kg^{-1}$)
h_{TC}	heat transfer coefficient ($W\,m^{-2}\,K^{-1}$)
$\dot{m}$	mass flow rate ($kg\,s^{-1}$)
m	mass (kg)
P	perimeter (m)
p	pressure (Pa)
$\dot{q}$	heat flux per unit area ($W\,m^{-2}$)
T	temperature (°C)
V	volume (m^3)
v	velocity ($m\,s^{-1}$)
α_k	void fraction $k = g$ for gas and $k = l$ for liquid
Δt	time step (s)
Δz	control volume length (m)
δ	rate of convergence
λ	thermal conductivity ($W\,m^{-1}\,K^{-1}$)
Γ	mass transfer rate per unit volume ($kg\,s^{-1}\,m^{-3}$)
ρ	density ($kg\,m^{-3}$)
τ_w	shear stress ($N\,m^{-2}$)
θ	inclination (rad)

Subscripts

g	gas or vapour
l	liquid
c, d	continuous (c) and dispersed (d) phases
e, w, n, s	east, west, north and south CV-faces
P	main grid node associated to main CVs
E, W, N, S	nearest east, west, north and south nodes associated to main grid node P
sat	saturation condition

to predict the thermal and fluid-dynamic behaviour of the fluid flow, looking for a realistic description of the phenomena.

Different one-dimensional mathematics have been used to describe the two-phase flow, from a simplified homogeneous equilibrium model (HEM) to a complex two-fluid model (TFM). The homogeneous model assumes the two-phase flow as a mixture. Therefore, non-interface exchange is considered, and only one pressure, one mixture velocity and one mixture temperature are used to describe the thermal and fluid-dynamic behaviour of the fluid flow (Levy, 1999; Morales-Ruiz et al., 2009). The separated two-phase flow model with no interface exchange assumes one pressure, different gas and liquid velocities, and equal gas and liquid temperature. In these kinds of models, the motion of the whole mixture is expressed by the mixture momentum equation. The relative motion between phases is taking into account by a kinematic constitutive equation. Thus, a simplified formulation can be written. The drift-flux model is within this model group (Wallis, 1969; Hibiki and Ishii, 2003). The two-fluid model proposes a separated model considering the interface exchange, different gas and liquid velocities, and two different gas and liquid temperatures. A group of conservative equations are proposed for each phase, forming a set of equations to obtain the pressure distribution, void fraction, and velocities and temperatures of the liquid and gas phases (Delhaye, 2001; Ishii and Hibiki, 2006).

All models commented above need a set of empirical information. Depending on the type of model and hypothesis assumed, the number of empirical expressions required can change. The two-fluid model require a significant number of empirical correlations to describe the different effects on the fluid flow along the interface, and between each phase and the wall. Most of these correlations have been obtained from nuclear systems literature (INEL, 2001; Bestion, 1990), e.g. heat transfer coefficient, shear stresses at the interface and at the wall, and some others specific parameters, i.e. the stratified angle. All of these parameters depend on the flow regime map, which is defined as function of the void fraction, the average mass flux and flow velocities.

In this work, two different flow pattern maps have been considered in function of the fluid. So, a flow regime map proposed to steam–water system (INEL, 2001) and a flow regime map proposed to boiling in refrigeration system (Wojtan et al., 2005) have been implemented. Specific information obtained from technical literature has been studied to define the most suitable empirical expression in function of flow regime and heat conditions (Ishii, 1984; Ullmann and Brauner, 2006; Issa et al., 2006).

Most of the studies published in the literature considering two-fluid models are focused on mathematical formulation (Delhaye et al., 1981; Yadigaroglu and Lahey, 1976), resolution methodology and numerical strategies (Tiselj and Petelin, 1997; Song and Ishii, 2001b; Toumi and Kumbaro, 1996), and instability problem due to the fact that the system of differential equations is non-hyperbolic (Song and Ishii, 2001a; Zhou and Podowski, 2001). A lot of these studies are always referring to nuclear systems and less attention has been devoted to refrigeration systems.

Some examples of application of the two-fluid models in the refrigeration field can be found in the next three papers. The first one is the study of a two-phase critical flow through a short tube orifice using a two-fluid model six equations (Liang and Chun-Lu, 2005). The six equations are formulated without taking into account the temporal term. These are discretized using the central difference scheme. The set of algebraic equations is solved by means of a step-by-step method following a sequential algorithm to find the length of the tube. The second paper explains the methodology followed to find the two-phase flow solution in double-pipe evaporators by means of the two-fluid model four equations (Sripattranpan et al., 2004). Applying the finite difference method, the equations are discretized and solved simultaneously using the Newton method of iteration. The third paper corresponds to the analysis of two-phase flow in double-pipe condensers (Wang, 1997). The two-phase flow is solved assuming non-equilibrium, and applying a two-fluid model four equations. A semi-implicit finite difference scheme is used to solve the governing equations. The set of non-linear equations system is solved by the Newton–Raphson method. Furthermore, the two-fluid model four equations has been used to study the stratified and slug two-phase flow inside horizontal tubes, together with the instabilities criteria (Issa and Kempf, 2003; Ansari and Shokri, 2007). The two-fluid model four equations has also been implemented by Bonizzi et al. (2009) to study the results dependence in function of the flow regime of near-horizontal gas–liquid flows in pipelines.

This paper is focused on detailed one-dimensional numerical simulation of phase change phenomena gas–liquid flow into pipes, assuming only one fluid. The basic field equation following the

transient two-fluid model consists of two continuity equations, two momentum equations and two energy equations, one equation for each phase. The six governing equations are discretized by means of the finite volume technique. These equations have been organized in a set of algebraic equations, which is solved using the three diagonal matrix algorithm TDMA. The coupling between momentum and continuity equations is solved with a semi-implicit pressure based method like SIMPLE (Patankar, 1980). This method is very usual and robust in the single-phase flow simulation. However, when two-phase flow occurs some convergence problems can appear during the iterative process. These problems occur due to the strong coupling between two momentum and two continuity equations, together with the effect of the interphase drag and mass transfer (Miller and Miller, 2003). The strong coupling makes the algorithm depend more on the velocities of previous solution iteration, producing instabilities along the simulation process. One way to solve this effect is writing explicitly the interfacial terms in continuity, momentum and energy equation and solve the set of equations to obtain a pseudo-solution which resolves the interphase coupling (Darwish and Moukalled, 2001).

Different numerical aspects have been evaluated with the aim of verifying the quality of the numerical solution. Transient and steady-state have been considered in this study and a group of four cases are presented. Three cases reported in the technical literature as benchmark cases are solved, and a comparison between numerical results and analytical solution is presented. Furthermore, the energy equation solution is tested with an in-tube evaporator case working under four different working conditions. The numerical results of this last case are compared with an experimental data obtained from a our experimental facility (Raush et al., 2009).

2. Governing equations

The mathematical formulation is based on the two-fluid model, which is an inter-penetrating model capable of defining the thermo-hydraulic behaviour of the fluid flow. The velocity, pressure, temperature and void fraction of each one of the phases are found by means of this method. The development of the field equations has been described by different authors in the technical literature, together with averaging processes to derive models suitable for computation (Ishii and Hibiki, 2006; Delhaye et al., 1981; Yadigaroglu and Lahey, 1976; Prosperetti and Tryggvason, 2007). The specific formulation to solve a one-dimensional two-phase flow by means of the two-fluid model is presented here. Six-equations are considered: mass, momentum and energy for each one of the phases. The formulation is focused on one-dimensional flows inside ducts or pipes of constant cross section. The analysis does not consider axial heat conduction or heat radiation.

The mass equations for each phase are expressed as:

$$\frac{\partial}{\partial t}(\alpha_l\rho_l)+\frac{\partial}{\partial z}(\alpha_l\rho_l v_l)=-\Gamma_g \tag{1}$$

$$\frac{\partial}{\partial t}(\alpha_g\rho_g)+\frac{\partial}{\partial z}(\alpha_g\rho_g v_g)=\Gamma_g \tag{2}$$

where α_g and α_l are the gas and liquid void fraction, v_l and v_g are the liquid and gas velocities, and Γ_g is the gas mass transfer rate per unit volume.

The momentum equations for each phase is expressed as:

$$\frac{\partial}{\partial t}(\alpha_l\rho_l v_l)+\frac{\partial}{\partial z}(\alpha_l\rho_l v_l^2)=-\alpha_l\frac{\partial p}{\partial z}-\frac{P_{wl}\tau_{wl}}{S}-\alpha_l\rho_l g\sin\theta-\frac{P_i\tau_{il}}{S}-\Gamma_g v_{il} \tag{3}$$

$$\frac{\partial}{\partial t}(\alpha_g\rho_g v_g)+\frac{\partial}{\partial z}(\alpha_g\rho_g v_g^2)=-\alpha_g\frac{\partial p}{\partial z}-\frac{P_{wg}\tau_{wg}}{S}-\alpha_g\rho_g g\sin\theta-\frac{P_i\tau_{ig}}{S}+\Gamma_g v_{ig} \tag{4}$$

where p is the total pressure, g is the gravitational acceleration, θ is the inclination angle of the pipe, τ_{wg} and τ_{wl} are the shear stresses acting on the gas and liquid phase at the wall, respectively, and τ_i is the shear stress between phases at the interface. The wall perimeter in contact with the liquid and gas is denoted by P_{wl} and P_{wg}, while S is the flux cross section area.

The energy equations expressed in terms of total enthalpy are:

$$\frac{\partial}{\partial t}(\alpha_l\rho_l h_l)+\frac{\partial}{\partial z}(\alpha_l\rho_l v_l h_l)=-\frac{\dot{q}_{wl}P_{wl}}{S}+\alpha_l\frac{\partial p}{\partial t}-\Gamma_g h_{il}+\frac{\dot{q}_{il}P_{il}}{S} \tag{5}$$

$$\frac{\partial}{\partial t}(\alpha_g\rho_g h_g)+\frac{\partial}{\partial z}(\alpha_g\rho_g v_g h_g)=-\frac{\dot{q}_{wg}P_{wg}}{S}+\alpha_g\frac{\partial p}{\partial t}+\Gamma_g h_{ig}+\frac{\dot{q}_{ig}P_{ig}}{S} \tag{6}$$

where $\dot{q}_{il}$ and $\dot{q}_{ig}$ are the heat transfer exchanged at the interface, while $\dot{q}_{wl}$ and $\dot{q}_{wg}$ are the heat transfered from the wall to the liquid and gas phase, respectively. These quantities are evaluated by means of empirical expressions.

3. Hyperbolic analysis of the governing equations

In the present paper, a one-dimensional single pressure six equations two-fluid model is used to modelize the thermal and fluid-dynamic behavior of the two-phase flow. It is well known that one-dimensional two-fluid model equations are ill-posed mathematically due to the loss of the hyperbolicity of the equations system under certain conditions (Issa and Kempf, 2003). A mathematical strategy is applied to verify the hyperbolic character of the equations system proposed in this paper.

The differential Eqs. (1)–(6) are written in matrix form for hyperbolic analysis as:

$$A\frac{\partial\Phi}{\partial t}+B\frac{\partial\Phi}{\partial z}=C \tag{7}$$

where A and B are coefficient matrices, Φ is the solution vector, which corresponds to $\Phi=[\alpha_l, v_l, v_g, p, h_l, h_g]^T$, and C is the source term, where the algebraic terms are included. The coefficient matrices A and B have the following form when all differential terms are expressed as function of the solution vector Φ:

$$\mathbf{A}=\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0 & 0\\ \rho_l v_l & \alpha_l\rho_l & 0 & 0 & 0 & 0\\ -\rho_g v_g & 0 & (1-\alpha_l)\rho_g & 0 & 0 & 0\\ \rho_l h_l & 0 & 0 & -\alpha_l & \alpha_l\rho_l & 0\\ \rho_g h_g & 0 & 0 & -(1-\alpha_l) & 0 & (1-\alpha_l)\rho_g \end{pmatrix}$$

and

$$\mathbf{B}=\begin{pmatrix} v_l & \alpha_l & 0 & 0 & 0 & 0\\ -v_g & 0 & (1-\alpha_l) & 0 & 0 & 0\\ \rho_l v_l^2 & 2\alpha_l\rho_l v_l & 0 & \alpha_l & 0 & 0\\ -\rho_g v_g^2 & 0 & 2(1-\alpha_l)\rho_g v_g & (1-\alpha_l) & 0 & 0\\ \rho_l v_l h_l & \alpha_l\rho_l h_l & 0 & 0 & \alpha_l\rho_l v_l & 0\\ -\rho_g v_g h_g & 0 & (1-\alpha_l)\rho_g h_g & 0 & 0 & (1-\alpha_l)\rho_g v_g \end{pmatrix}$$

The characteristic value of the equations set are obtained by solving the eigenvalue system. An algebraic expression is obtained after applying $f(\lambda)=det[A\lambda-B]=0$, where λ is the system characteristic value:

$$f(\lambda)=-\alpha_l\alpha_g[\alpha_g\rho_l(\lambda-v_l)^2+\alpha_l\rho_g(\lambda-v_g)^2][\alpha_l\rho_l(\lambda-v_l)]\times[\alpha_g\rho_g(\lambda-v_g)] \tag{8}$$

The solution of the fourth-order polynomial defined above gives the roots or the characteristic values of the system of equations. The solution gives that real eigenvalues are obtained only for an specific case, when the phasic velocities are equals ($u_g = u_l$).

$$\begin{cases} \lambda_1 = v_l \\ \lambda_2 = v_g \\ \lambda_{3,4} = \frac{\alpha_g\rho_l v_l + \alpha_l\rho_g v_g}{\alpha_g\rho_l + \alpha_l\rho_g} \pm \frac{\sqrt{-\alpha_l\rho_l\alpha_g\rho_g(v_g - v_l)^2}}{\alpha_g\rho_l + \alpha_l\rho_g} \end{cases} \quad (9)$$

The model proposed in this paper has not a complete set of real eigenvalues, which indicates that the model does not represent a well-posed equations system. In that sense, special mechanisms should be applied in order to obtain stable numerical results. One of them is based on the use of staggered grid and donor cell techniques, which gives a sufficient amount of numerical diffusion to obtain stable numerical results for many transient two-phase flow conditions (Stäadtke, 2006).

On the other hand, a well-posed system of the differential equations for the one-dimensional two-fluid model can be achieved including additional terms, such as virtual mass force, surface tension force, or using the hydrodynamic pressure rather than the hydrostatic pressure (Hwang, 2003; Ansari, 2004). Even so, the applicable range of their solutions is limited (Song and Ishii, 2001a). In our case, no additional terms are considered in the resolution of equations system.

4. Empirical information

Frictional coefficients, heat transfer coefficients and interfacial area are calculated by means of empirical correlations. A good evaluation of these factors depends on the definition of the flow regime, which indicates how the phase distributions affect the physical nature of the system. A flow pattern map is a tool which indicates the visible distribution or structure of the phase into the pipe. It depends on the work fluid. Some parameters (i.e. type of fluid, flow geometry, inclination angle, velocities and boundary conditions) are necessary information in the definition of the flow regime from a flow pattern map.

A flow pattern map has to be considered depending on the kind of fluid used: water-vapour (INEL, 2001); a related group of fluid refrigerants (Kattan et al., 1998; Wojtan et al., 2005); and carbone dioxide (Cheng et al., 2006) among others. In general, a flow pattern map defines regions or areas where stratified, bubbly, slug, annular and dispersed regimes appear. The regime definition is a function of the void fraction and the relative velocity between phases in case of water–vapour fluid; and the mass fraction, the average mass flux and the heat flux applied in case of refrigerants fluid. As an example, a flow pattern map used in the numerical resolution method proposed here is depicted in Fig. 1, which corresponds to a flow regime map characteristic of one of cases presented in Section 7.4.

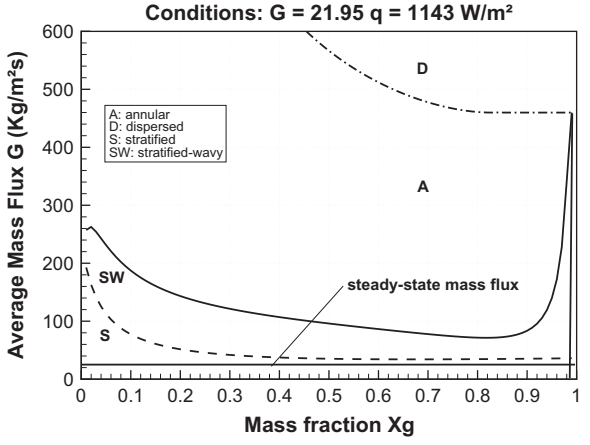


Fig. 1. Flow pattern map according to boiling refrigerants Kattan et al. (1998) and Wojtan et al. (2005).

A specific critical velocity criteria proposed by INEL (2001) is considered in all test cases, in order to know if the transition between regimes affects the numerical results stability. Following expression has been used to evaluate the critical velocity value as a function of the position:

$$v_{crit} = \begin{cases} 0.5\left[\frac{(\rho_l - \rho_g)g\alpha_g S}{\rho_g D \sin\varphi}\right]^{1/2}(1.0 - \cos\varphi) & \text{horizontal flow} \\ 0.35\left[\frac{(\rho_l - \rho_g)gD}{\rho_l}\right] & \text{vertical flow} \end{cases} \quad (10)$$

where φ is the angle between the vertical axis and the liquid level, which is related to the void fraction by means of $\alpha_g\pi = \varphi - \sin\varphi\cos\varphi$.

Once the kind of flow pattern map to be used in function of the fluid is defined, a short summary of the empirical expressions used in this work depending on the flow regime are listed below.

The momentum equations require the evaluation of the frictional shear stress at the wall and at the gas–liquid interface. The interfacial friction is evaluated by means of the drag coefficient method proposed in RELAP5 code (INEL, 2001):

$$\tau_{ig} = -\tau_{il} = \frac{1}{2}C_D\rho_g|v_g - v_l|(v_g - v_l) \quad (11)$$

where the interfacial friction factor C_D is defined as the maximum value between $64/Re_i$, general expression for laminar flow, and $0.3164/(Re_i)^{1/4}$, Blasius correlation for turbulent flow. The interfacial Reynolds number is defined as $Re_i = D_i\rho_g|v_g - v_l|/\mu_g$, where D_i is the equivalent wetted diameter, $D_i = \alpha\pi D/(\omega + \sin\omega)$. The angle ω represents the angle between the vertical symmetry axis and the intersection of the liquid level, this value is used to calculate the interfacial area as $A_i = 4\sin\omega/(\pi D)$.

Wall shear effects are evaluated considering both the Lockhart–Martinelli model and the Heat Transfer and Fluid Flow Service (HTFS)-modified Baroczy correlation (INEL, 2001). According to the standard procedure, the pressure drop of two-phase flow is calculated from the pressure drop evaluated assuming only one phase, liquid or gas, and correcting the obtained value with the Lockhart–Martinelli two-phase multiplier Φ_l^2 or Φ_g^2 evaluated with HTFS correlation.

$$\left(\frac{dp}{dz}\right)_{TP} = \Phi_l^2\left(\frac{dp}{dz}\right)_l = \Phi_g^2\left(\frac{dp}{dz}\right)_g \quad \text{and} \quad \left(\frac{dp}{dz}\right)_k = f_k\frac{G^2}{2\rho_k D}$$
$$\text{with:} \quad \Phi_g^2 = \chi^2 + C\chi + 1 \quad \text{and} \quad \Phi_l^2 = 1 + C/\chi + 1/\chi^2 \quad (12)$$

where k represents the liquid or gas phase flowing alone, f_k is the friction factor evaluated as function of Reynold number, C is the correlation coefficient evaluated as function of mass flux G, and χ is the Lockhart–Martinelli ratio defined as: $\chi = (dp/dz)_l/(dp/dz)_g$.

Based on the pressure drop in the two-phase obtained above, it is possible to define a partitioning of the overall wall friction between phases, which is independent of the model for interfacial friction. The individual wall shear of the liquid and gas phase can be defined from the overall wall friction, the void fraction and a parameter Z^2 in the following way (INEL, 2001):

$$\tau_{wg} = \alpha_g\left(\frac{dp}{dz}\right)_{TP}\left(\frac{1}{\alpha_g + \alpha_l Z^2}\right)\frac{S}{P_{wg}}$$
$$\tau_{wl} = \alpha_l\left(\frac{dp}{dz}\right)_{TP}\left(\frac{Z^2}{\alpha_g + \alpha_l Z^2}\right)\frac{S}{P_{wl}} \quad (13)$$
$$Z^2 = \frac{f_l\rho_l v_l^2\frac{\alpha_{wl}}{\alpha_l}}{f_g\rho_g v_g^2\frac{\alpha_{wg}}{\alpha_g}}$$

where $\alpha_{wg} = \omega/\pi$ and $\alpha_{wl} = 1 - \alpha_{wg}$ representing the gas and liquid fraction on the wall, and defined as a function of the angle ω in stratified flow regime.

The mass flux per unit volume transfered through the interface can be defined from a heat balance at the interface. The sum of the mass transfered at the interface Γ_i due to the energy exchanged between phases, and the mass transfer from the wall to interface Γ_w due to the external heat transfer can be expressed as (Levy, 1999 and INEL, 2001):

$$\begin{aligned} \Gamma_w &= \frac{\dot{q}_{wl} P_{wl}}{(h_{g,sat} - h_l)} \\ \Gamma_i &= -\frac{\left[\left(\frac{\dot{q}_{il} P_{il}}{S}\right) - \Gamma_w (h_{g,sat} - h_l) + \left(\frac{\dot{q}_{ig} P_{ig}}{S}\right)\right]}{(h_{g,sat} - h_l)} \\ \Gamma &= \Gamma_w + \Gamma_i \quad \text{where} \quad \Gamma_g = \Gamma \quad \text{and} \quad \Gamma_l = -\Gamma \end{aligned} \tag{14}$$

The heat transferred per unit of volume at the interface is evaluated in the following form:

$$\begin{aligned} \left[\frac{\dot{q}_{il} P_{il}}{S}\right] &= h_{TC,il}(T_{sat} - T_l) + \Gamma_w (h_{g,sat} - h_l) \\ \left[\frac{\dot{q}_{ig} P_{ig}}{S}\right] &= h_{TC,ig}(T_{sat} - T_g) \end{aligned} \tag{15}$$

The interfacial heat transfer coefficients per unit volume, $h_{TC,il}$ and $h_{TC,ig}$, for the liquid and the gas phase respectively, have been evaluated from the empirical expressions proposed in RELAP5 manual code (INEL, 2001). These expressions consist of two parts. The first one is based on Dittus–Boelter correlation, which is a well-known expression to evaluate the heat transfer coefficient in single-phase. The second part is a term whose value is a large number. This is a strategy to avoid non-metastable conditions in each one of the phases.

$$h_{TC,il} = \begin{cases} \frac{\lambda_l}{D_l}\left[\left(0.023 Re_l^{0.8} F_{12}\right) - 3.81972 \frac{(T_{sat} - T_l)\rho_l c p_l}{[\rho_g h_{lg} max(4\alpha, 1)]}\right] A_i & \text{for } (T_{sat} < T_l) \\ \frac{\lambda_l}{D_l}\left(0.023 Re_l^{0.8}\right) A_i & \text{otherwise} \\ \text{with } F_{12} = 1 + \varepsilon(250 + 50\varepsilon) \quad \varepsilon = max[0, -(T_{sat} - T_l)] & \end{cases}$$

$$h_{TC,ig} = \begin{cases} \frac{\lambda_g}{D_g}\left[\left(0.023 Re_g^{0.8}\right) + (4e^5 F \max[0, (0.25 - \alpha)])\right] A_i & \text{for } (T_{sat} < T_g) \\ 0.0 & \text{otherwise} \\ \text{with } F = 1 + \eta(100 + 25\eta) \quad \eta = |max[-2, -(T_{sat} - T_g)]| & \end{cases} \tag{16}$$

where F_{12} increases the heat transfer coefficient value with the aim of maintain a metastable condition returning liquid temperature toward saturation temperature. The same strategy is followed in case of the gas phase, the parameter F increases the heat transfer coefficient to get the gas temperature quickly toward the saturation temperature.

The heat transfer at the wall is calculated using the difference between wall and fluid phase temperature, the superficial area in contact with the phase, and the heat transfer coefficient of the phase. The heat transfer coefficients of the liquid at the wall $h_{TC,wl}$ are evaluated considering the contribution of the nucleate pool boiling, $h_{TC,nb}$, and the convective liquid flow boiling, $h_{TC,cb}$, i.e. (Wattelet et al., 1994):

$$\begin{aligned} h_{TC,cb} &= \frac{\lambda_l}{D} 0.0133 Re_l^{0.69} Pr_l^{0.4} \\ h_{TC,nb} &= 55 P_r^{0.12} [-log_{10} P_r]^{-0.55} M^{-0.5} q^{0.67} \\ h_{TC,wl} &= [h_{TC,cb}^3 + h_{TC,nb}^3]^{1/3} \end{aligned} \tag{17}$$

where Re_l is the Reynolds number based on liquid and Pr_l is the Prandtl number of liquid, P_r the reduced pressure, M the fluid molecular weight, and q the heat flux applied.

Following the Frepoli suggestion (Frepoli, 2001), the heat transfer coefficient of the gas at the wall is evaluated considering the correlations by Wong–Hochreiter and Dittus–Bolter:

$$h_{TC,wg} = \frac{\lambda_g}{D}\left[max\left(0.0797 Re_g^{0.6774} Pr_g^{0.333}, 0.023 Re_g^{0.8} Pr_g^{0.4}\right)\right] \tag{18}$$

The heat transfer coefficient in the post dry-out zone has been evaluated by means of the critical heat flux look-up tables method proposed by Groeneveld (Groeneveld et al., 1996).

5. Numerical aspects and discretization

The one-dimensional governing Eqs. (1)–(6) can be integrated in terms of the local averaged fluid variables using the finite control volume technique. The concept of a staggered mesh has been used, where velocities are defined at the cell boundaries, while pressure, void fraction and enthalpies are located at the middle of the main volume. A schematic representation of the control volume, together with the definition of the nodes are depicted in Fig. 2.

Transient terms are evaluated using a first-order backward differencing scheme, while convective terms at the control volume faces are evaluated applying a first-order, a second-order or a higher-order numerical scheme. The nomenclature proposed by Prosperetti and Tryggvason (2007) is used to implement different numerical schemes.

The value of the variable ϕ (enthalpy, void fraction and pressure) at the face f are calculated by means of the expression: $\phi_f = \phi_c + \frac{1}{2} l_f (\phi_d - \phi_c)$, where ϕ_c represents the value of the variable at the current node, ϕ_d is the value at the node immediately downstream and ϕ_u the value of the variable at the node immediately upstream of the current node, see Fig. 2. The limiter l_f value depends on the numerical scheme: (i) $l_f = 0$ for upwind scheme; (ii) $l_f = 1$ for central-difference scheme or (iii) $l_f = max[0, min[4\delta, 1/4(3 + \delta), 2]]$ for smart scheme, where $\delta = (\phi_c - \phi_u)/(\phi_d - \phi_c)$. The influence of these numerical schemes on the stability and numerical results has been analyzed in some of the cases studied, taking into account the conclusions cited by Liao et al. (2008).

Darwish and Moukalled (2001) presented and grouped the multi-flow segregated algorithms under two categories: the geometric conservation based algorithms (GCBA) and the mass conservation based algorithms (MCBA). Both algorithms are extensions of

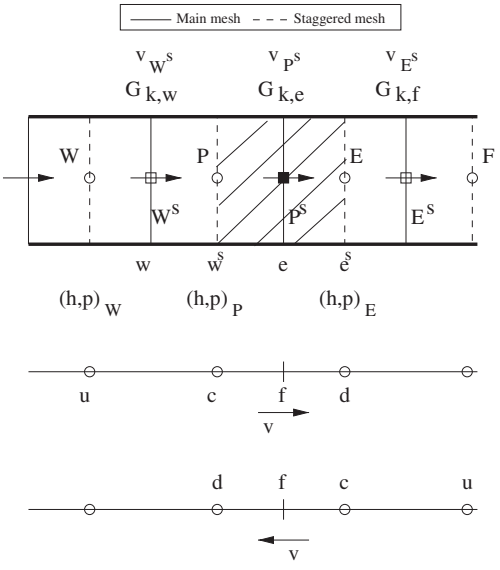


Fig. 2. Schematic control volumes.

single-fluid segregated algorithms. A MCBA algorithm has been used in this paper to solve the two-phase flow inside pipes. The main reason to choose a MCBA algorithm is that it scales down automatically to handle single-fluid, as it has been derived directly from single-fluid algorithm. The GCBA algorithm needs some modifications in the mathematical formulation. The poor conservation of the lighter fluid in MCBA algorithms has been solved by means of normalizing the continuity equations, using a weighting factor, i.e. a reference density of the phase $\rho_{k,ref}$.

The semi-implicit scheme is used in the discretization of the governing equations. As a consequence, the time step definition is not dependent of a CFL time constraint. This is an advantage in comparison with the explicit schemes, in which CFL time constraint is a requirement. The time step has not any computational limitations, and it is limited only by the phenomenon of the case.

Momentum equations are discretized on the staggered CVs. After subtracting the mass conservation equation multiplied by the velocity phase v_{k,P^s}, an algebraic expression is obtained for each one of the phases.

$$\begin{aligned}&\frac{1}{\Delta t}\left[\frac{(\rho^o_{k,P}\alpha^o_{k,P}+\rho^o_{k,E}\alpha^o_{k,E})}{2}\right](v_{k,P^s}-v^o_{k,P^s})+\frac{1}{\Delta z}\left[G_{k,e^s}(v_{k,e^s}-v_{k,P^s})\right.\\&\left.-G_{k,w^s}(v_{g,w^s}-v_{k,P^s})\right]=-\frac{1}{\Delta z}\frac{(\alpha_{k,P}+\alpha_{k,E})}{2}[p_E-p_P]\\&-g\frac{(\rho_{k,P}\alpha_{k,P}+\rho_{k,E}\alpha_{k,E})}{2}sin\,\theta-\frac{P_{wk}\tau_{wk}}{S}-\frac{P_i\tau_{ik}}{S}+\Gamma_k[v_{ik,P^s}-v_{k,P^s}]\end{aligned}\tag{19}$$

where terms v_{k,e^s} and v_{k,w^s} are evaluated in function of the chosen numerical scheme. The mass flow rate per unit of area is defined by $G_{k,e}=(\alpha_{k,e}\rho_{k,e}v_{k,e})$, where k means the gas or liquid phases, and e means the CVs face. The void fraction is defined as the relation between the cross area occupied by the gas and the total cross area, $\alpha_k = S_k/S$. The interface velocities v_{il} and v_{ig} are assumed to be equal and are defined as $v_{il}=v_{ig}=\psi v_g+(1-\psi)v_l$, where $\psi=0$ if $\Gamma\geqslant 0$ and $\psi=1$ if $\Gamma<0$, being Γ the mass transfer rate per unit volume (INEL, 2001).

A pressure correction equation is obtained adding the gas and liquid discretized continuity equations, and normalizing it with liquid and gas reference densities, $\rho_{k,ref}$. Following the mass conservation-based algorithm (MCBA) (Darwish and Moukalled, 2001):

$$\begin{aligned}&\frac{\Delta z}{\Delta t\rho_{g,ref}}\left[\rho^*_{g,P}\alpha^*_{g,P}+\frac{\rho^*_{g,P}\alpha^*_{g,P}}{\gamma p^*_P}p'_P-\rho^o_{g,P}\alpha^o_{g,P}\right]\\&+\frac{\Delta z}{\Delta t\rho_{l,ref}}\left[\rho^*_{l,P}\alpha^*_{l,P}+\frac{\rho^*_{l,P}\alpha^*_{l,P}}{\gamma p^*_P}p'_P-\rho^o_{l,P}\alpha^o_{l,P}\right]+\frac{1}{\rho_{g,ref}}\left[G^*_{g,e}+G^*_{g,e}\frac{p'_e}{\gamma p^*_e}\right.\\&+\left.\left[\rho_{g,e}\alpha_{g,e}\right]^*d_e^{v_{g,P^s}}(p'_P-p'_E)-G^*_{g,w}-G^*_{g,w}\frac{p'_w}{\gamma p^*_w}\right.\\&-\left.\left[\rho_{g,w}\alpha_{g,w}\right]^*d_e^{v_{g,W^s}}(p'_W-p'_P)\right]+\frac{1}{\rho_{l,ref}}\left[G^*_{l,e}+G^*_{l,e}\frac{p'_e}{\gamma p^*_e}\right.\\&+\left.\left[\rho_{l,e}\alpha_{l,e}\right]^*d_e^{v_{l,P^s}}(p'_P-p'_E)-G^*_{l,w}-G^*_{l,w}\frac{p'_w}{\gamma p^*_w}\right.\\&-\left.\left[\rho_{l,w}\alpha_{l,w}\right]^*d_e^{v_{l,W^s}}(p'_W-p'_P)\right]=0\end{aligned}$$

where

$$\begin{aligned}&G_{g,e}=(\rho_{g,e}\alpha_{g,e})v_{g,e}=(\rho_{g,e}\alpha_{g,e})^*v^*_{g,e}+(\rho_{g,e}\alpha_{g,e})'v^*_{g,e}+(\rho_{g,e}\alpha_{g,e})^*v'_{g,e}\\&G^*_{g,e}=\left(\rho^*_{g,e}\alpha^*_{g,e}\right)v^*_{g,e}\\&v'_{g,e}=v'_{g,P^s}=d_e^{v_{g,ps}}(p'_P-p'_E)\\&(\rho_{g,e}\alpha_{g,e})'=\frac{1}{\rho_{g,ref}}p'_e=\frac{1}{a^2}p'_e=\frac{1}{\gamma rT_e}p'_e=\frac{\rho^*_{g,e}\alpha^*_{g,e}}{\gamma p^*_e}p'_e\\&(\rho_{g,e}\alpha_{g,e})'v^*_{g,e}=G^*_{g,e}\frac{1}{\gamma p^*_e}p'_e\end{aligned}\tag{20}$$

The superscript * indicates the guess value of the considered variable, while the superscript $'$ indicates a correction value. The terms $\frac{p'_e}{\gamma p^*_e}$ and $\frac{p'_w}{\gamma p^*_w}$ depend on the numerical scheme used.

The gas and liquid void fraction are obtained from the continuity equations of each phase evaluated on the main mesh. The discretized equation is expressed as:

$$\frac{\rho_{k,P}\alpha_{k,P}}{\Delta t}+\frac{1}{\Delta z}[(\rho_{k,e}v_{k,e})\alpha_{k,e}-(\rho_{k,w}v_{k,w})\alpha_{k,w}]=\Gamma_k+\frac{\rho^o_{k,P}\alpha^o_{k,P}}{\Delta t}\tag{21}$$

where terms $\alpha_{k,e}$ and $\alpha_{k,w}$ are evaluated in function of the numerical scheme applied. Due to numerical errors, the values obtained after solving the last equation for each phase may not satisfy the geometric condition $\alpha_g+\alpha_l=1$. This fact can produce problems in the convergence process. A way to overcome this problem is to re-evaluate the volume fraction values obtained before, α^f_k, using the following expression:

$$\alpha_k=\frac{\alpha^f_k}{\alpha^f_g+\alpha^f_l}\tag{22}$$

The gas and liquid energy equations have been obtained discretizing on the main mesh. The difference between the discretized energy equation and the continuity equation multiplied by the value of enthalpy $h_{k,P}$ in the center at the CV is expressed as:

$$\begin{aligned}&\frac{\rho^o_{k,P}\alpha^o_{k,P}h_{k,P}}{\Delta t}-\frac{\rho^o_{k,P}\alpha^o_{k,P}h^o_{k,P}}{\Delta t}+\frac{1}{S\Delta z}[G_{g,e}S_e(h_{k,e}-h_{k,P})\\&-G_{k,w}S_w(h_{k,w}-h_{k,P})]=\left[\frac{q_{ik}P_{ik}}{S}\right]_P+\left[\frac{q_{wk}P_{wk}}{S}\right]_P+\Gamma_k[h_{ik}-h_{k,P}]\\&+\alpha_{k,P}\frac{(p-p^o)_P}{\Delta t}\end{aligned}\tag{23}$$

where terms $h_{k,e}$ and $h_{k,w}$ are evaluated according to the selected numerical scheme. The gas and liquid interfacial enthalpies, h_{ig} and h_{il}, are assumed to be equal to the gas and liquid saturated enthalpy.

6. Numerical resolution

The discretized equations described above have been organized in a generic discretized form $a_P\phi_P=a_E\phi_E+a_W\phi_W+b_P$. The set of algebraic equations is solved using the well-known three diagonal matrix algorithm TDMA. The coupling between momentum and continuity is solved by means of the semi-implicit pressure based method SIMPLE (Patankar, 1980), which is a quite robust technique. Although this method has been commonly used in single-phase flow, some problems can appear along the convergence process in case of the two-phase flows. It is a consequence of an extra coupling generated at the interface, due to the relative velocities term in each one of the momentum equations. One way to solve this problem is performing an explicit evaluation of this term. Other possibilities can also be applied, e.g. the partial elimination algorithm PEA, or the simultaneous solution of non-linearly coupled equations SINCE (Darwish and Moukalled, 2001).

The global algorithm consists of solving implicitly velocities, v_k, using guess pressure p^* and density $\rho_k{}^*$ fields. The pressure correction equation is solved to obtain the pressure correction value p'. After that, velocities and pressure are updated: $v_k=v^*_k+v'_k$, $p=p^*+p'$. The void fraction values α_k are solved, and a re-evaluation of it should be carried out to take into account the fact that the sum of gas and liquid void fraction will be 1. Finally, the energy equations in terms of enthalpy, h_k, are solved using the latest values of the different variables. An iterative process should be made until the convergence criteria is reached. The comparison between the maximum local residual value of each one of the variables, ϕ, and the error criteria allowed, δ, is used to assure that results have converged, i.e. *max* (residual value of ϕ) $<\delta$. The process runs

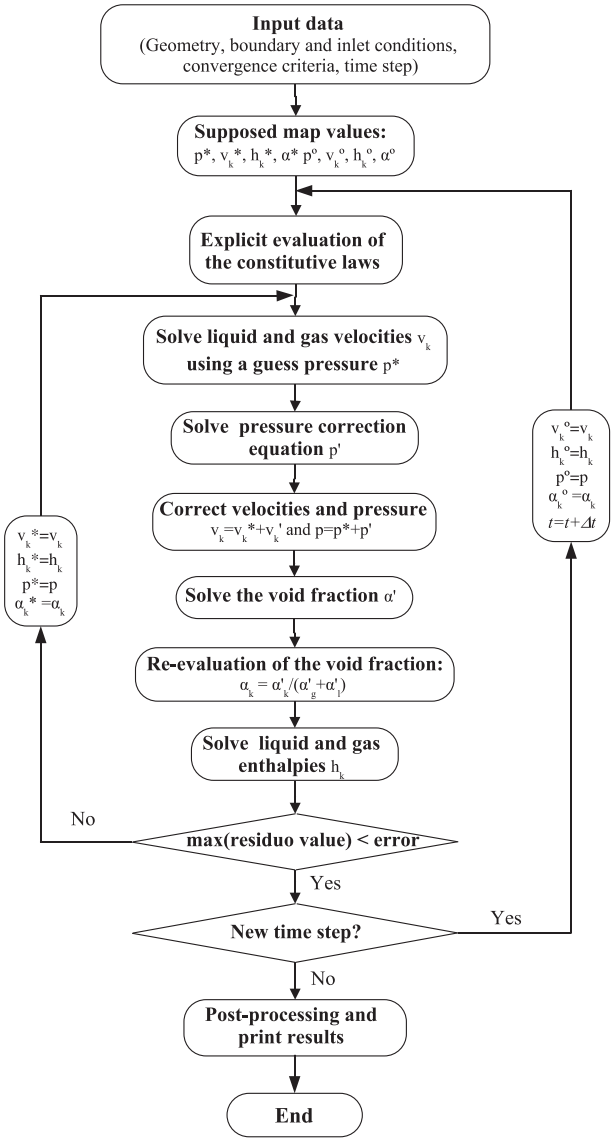


Fig. 3. Algorithm to solve the two-phase flow using a pressure based method.

step-by-step in the time direction until some criteria is satisfied (e.g. steady-state conditions are reached, or just after some physical period of time). An schematic representation of the algorithm is depicted in Fig. 3.

The convergence of the numerical results depends on two main aspects: the mesh size and the error criteria choice. The numerical predictions become grid-independent in function of the mesh size. Therefore, when the number of CVs is sufficiently increased, the numerical results are independent from this parameter. A similar behaviour can be see in function on the error criteria. If the error criteria is reduced enough, the numerical results will be independent from this parameter. The influence of both parameters is shown in the Water Faucet case in the next section.

7. Results

The numerical solution of the two-phase flow phenomena inside pipes is analyzed by means of the model proposed in previous section. Different test cases have been used to test the numerical procedure and the quality of the numerical results in comparison with analytical expressions obtained from the technical literature, together with experimental data obtained from an experiment setup (Raush et al., 2009).

7.1. Water Faucet

The first case is a well-known water faucet problem (Hewitt, 1983). This is a transient case reported as benchmark proof by many researchers in two-fluid models. It consists of a 12 m length and 1 m diameter vertical pipe. The pipe is assumed to be adiabatic, and is filled with water at the following inlet fluid conditions: gas void fraction 0.2, gas velocity 0.0 m/s, liquid velocity 10 m/s, temperature 50 °C, and outlet pressure 10^5 Pa. An schematic representation of this case is depicted in Fig. 4.

The water faucet assumes that frictional forces and thermal effects on interface are negligible. This case shows gravity effects over fluid acceleration. Illustrative results of this transient case are depicted in Fig. 5, where evolution of the gas void fraction and both liquid and gas velocities are shown at different times.

In agreement with the boundary conditions, an unconditionally well-posed of the equations system is not achieved due to the fact that liquid and gas velocities are not equals (Ansari and Shokri, 2011). In that sense, the numerical result shows an instability that grown when the mesh size is refined or Δz value is reduced.

The mesh size influence on the solution has been studied using a first order Upwind scheme. Five different uniform mesh size were used: 20 CVs, 120 CVs, 600 CVs, 1200 CVs and 3000 CVs, with a value of 10^{-8} as convergence criteria and a time state of 0.0005 s. A comparison between numerical results and analytical solution at instant 0.5 s are depicted in Fig. 6 for the void fraction and liquid velocity at different z positions.

The values of gas void fraction and liquid velocity at a given section (z = 5 m), and at the same instant (t = 0.5 s), are given in Table 1 using Upwind scheme and considering different meshes and convergence criteria. The analysis shows that, when the mesh is refined, the agreement between numerical and analytical solution is improved. The difference between numerical results obtained with 600 CVs and 3000 CVs reported errors of 5.12% and 4.4% in the gas void fraction, respectively. Unfortunately, some damping starts in the numerical solution when the mesh size is more refined, indicating that instable and inaccurate results can be obtained. Due to the fluctuations in the numerical results when the size mesh is higher than 1200 CVs, it is not possible to find an independent solution of the mesh size for this case.

Regarding convergence criteria error, a mesh of 1200 CVs has been used in all results to show the difference produced by the error criteria choice. After an error criteria of 10^{-8} the difference is very small in void fraction and liquid velocity values.

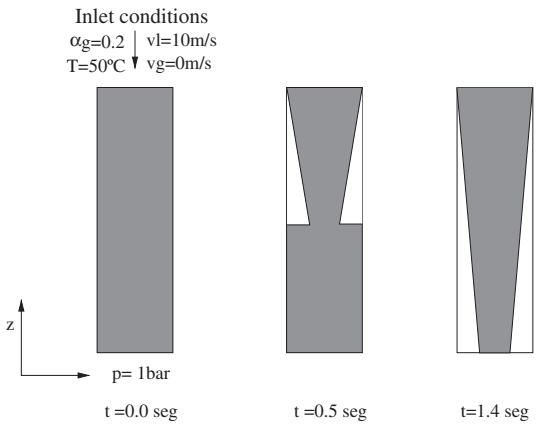


Fig. 4. Schematic representation of the Water Faucet problem along the time.

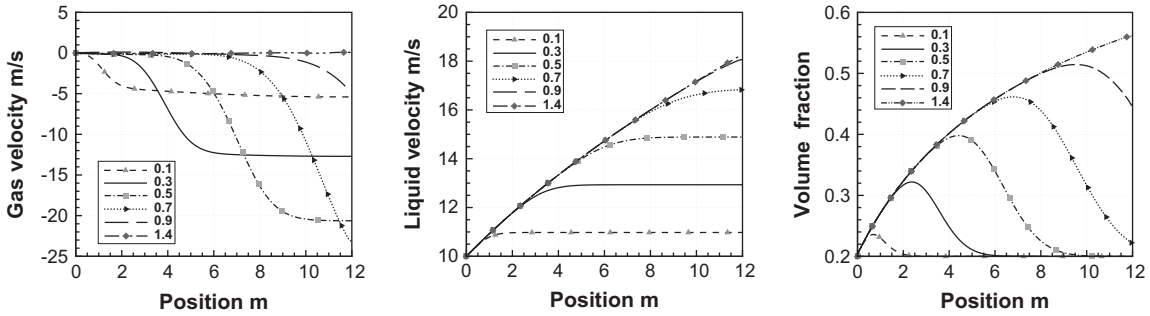


Fig. 5. Water Faucet results along the time in seconds.

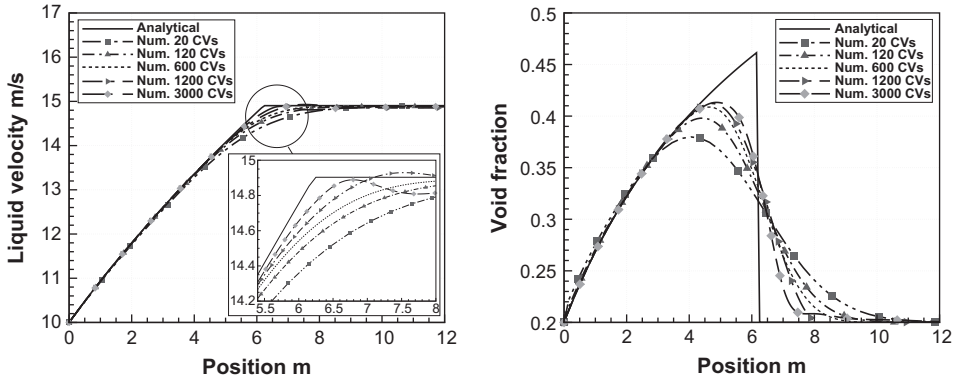


Fig. 6. Analytical and numerical comparison at $t = 0.5$ s of Water Faucet case.

Table 1
Mesh size and error criteria influence on Water Faucet case: (i) using a convergence criteria of 10^{-8} the mesh size influence is detailed at $t = 0.5$ s and $z = 5$ m and (ii) using 1200 CVs the convergence criteria influence is shown at $t = 0.5$ s and $z = 5$ m.

Mesh	α_g	v_l (m/s)
20	0.365918	13.8995
60	0.386305	13.9527
120	0.389478	13.9915
600	0.404425	14.0609
1200	0.409692	14.0561
1800	0.410150	14.0430
3000	0.412854	14.0462
Analytical	0.431910	14.1025
Convergence criteria	α_g	v_l (m/s)
10^{-5}	0.416405	14.0908
10^{-6}	0.412930	14.0705
10^{-7}	0.409251	14.0561
10^{-8}	0.409692	14.0561
10^{-9}	0.406551	14.0452
10^{-10}	0.406552	14.0452

The evaluation of the convective terms using different numerical schemes and their influence on the numerical results were analyzed in this case. Numerical results of liquid velocity and void fraction using meshes of 60 CVs and 600 CVs, and

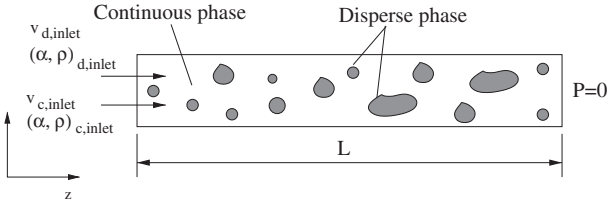


Fig. 7. Schematic representation of the horizontal tube with two phases flowing inside.

applying Upwind, CDS and Smart schemes are shown in Table 2. A time step of 0.0005 s and an accuracy criteria of 10^{-8} have been used to obtain the numerical results of each one of the numerical schemes applied. The differences between numerical results and the analytical solution are shown for each one of the schemes tested. The Smart scheme gives the better results than the other two schemes. However, if the mesh size is refined, the difference between the numerical results obtained using any of the three schemes is drastically reduced. Obviously, until the instability described above appears.

The numerical results obtained in this paper are in agreement with the numerical results of the study carried out by Ansari and

Table 2
Numerical scheme influence on Water Faucet case results at $t = 0.5$ s and $z = 5$ m.

Mesh size	60 CV				600 CV			
Scheme	α_g	Error %	v_l (m/s)	Error %	α_g	Error %	v_l (m/s)	Error %
Upwind	0.386305	10.56	13.9527	0.92	0.404425	6.36	14.0609	0.29
CDS	0.397963	7.86	13.9518	0.92	0.407440	5.66	14.0608	0.29
Smart	0.401817	6.77	14.0866	0.03	0.404403	6.37	14.0607	0.29

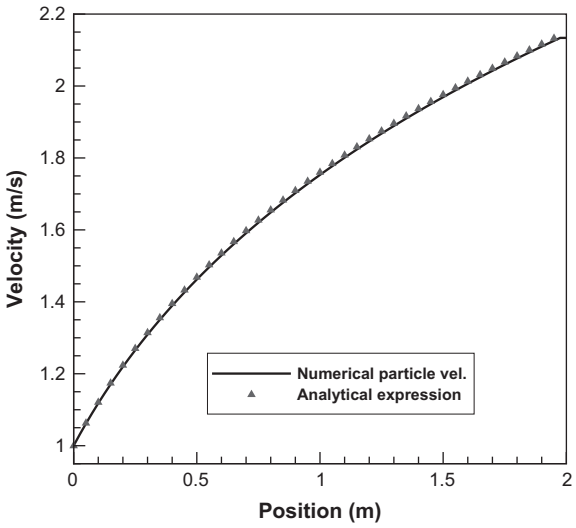


Fig. 8. Dilute gas–solid flow: comparison between numerical and analytical particle velocity along the tube.

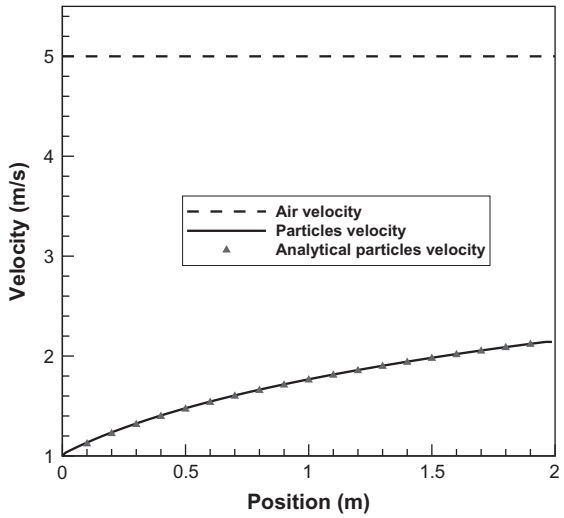


Fig. 9. Velocity distribution of the dense gas–solid flow.

Shokri (2011), in which the instability characteristic of the water faucet case is also analyzed.

7.2. Horizontal particle transport

A schematic representation of the physical situation is shown in Fig. 7. Two different cases are considered. The first case is based on the steady flow of solid particles (dispersed phase) suspended in a free stream of air (continuous phase). The second case corresponds to a steady flow of air bubbles (dispersed phase) suspended in a free stream of water (continuous phase). These cases show the influence of the interfacial drag force in the fluid dynamic behaviour of the phases. Other forces, such as gravity or inter-particle forces, are neglected.

The interfacial drag force of the disperse and the continuous phases are evaluated as (Moukalled and Darwish, 2002):

$$\frac{P_i\tau_{i,d}}{S}=\frac{3}{8}\frac{C_D}{r_p}\alpha_d\rho_c|v_d-v_c|(v_d-v_c)\quad\text{and}\quad\frac{P_i\tau_{i,c}}{S}=-\frac{P_i\tau_{i,d}}{S}\tag{24}$$

where subindex c and d indicate dispersed and continuous phase; a drag coefficient of $C_D = 0.44$ is used.

The first case is a dilute gas–solid flow, in which solid particles suspended in a free stream of gas are flowing inside a horizontal tube. The physical properties of the two-phases are: void fraction of the dilute particles $\alpha_d = 10^{-5}$, particle radius $r_p = 1$ mm, and density relation between dilute particles and continuous air stream $\rho_d/\rho_c = 2000$. The initial velocity of the dilute particles is 1 m/s, while the air velocity is 5 m/s.

The dilute particle velocity can be evaluated from an analytical solution, which is expressed as function of the position z, the inlet conditions, and the properties of both phases as (Moukalled and Darwish, 2002):

$$ln[v_{c,inlet}-v_d]+\frac{v_{c,inlet}}{v_{c,inlet}-v_d}=\frac{3}{8}\frac{\rho_c}{\rho_d}\frac{C_D}{r_p}z+ln[v_{c,inlet}-v_{d,inlet}]+\frac{v_{c,inlet}}{v_{c,inlet}-v_{d,inlet}}\tag{25}$$

The numerical results have been obtained using a mesh size of 100 CVs, a time step of 0.001 s and an error criteria of 10^{-6}. A comparison between numerical results and analytical data is shown in Fig. 8, where a good agreement between values is shown. The air stream is not affected by the dilute concentration of particles, while the dilute particles stream is accelerated by the action of the air stream.

A variation of the first case is the dense gas–solid flow, the particle concentration is $\alpha_{d,inlet} = 10^{-2}$. Although the dispersed phase void fraction value is low, the density relation between dispersed and continuous phases is large: $\alpha_d\rho_d/\alpha_c\rho_c = 20$. Thus, the disperse phase carries most of the inertia of the mixture. The solid particles velocity suffers an increase, while the air velocity is slightly

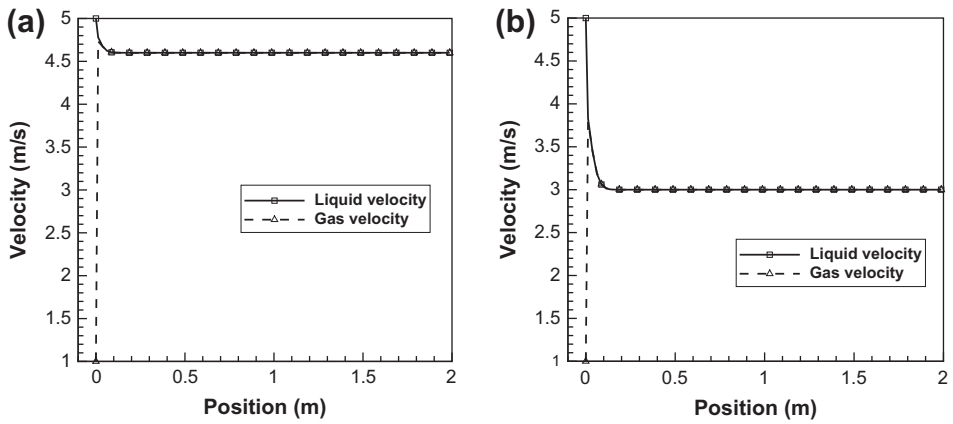


Fig. 10. Gas and liquid velocities: (a) dilute bubbly flow and (b) dense bubbly flow.

reduced. The verification of this phenomena can be highlighted using the equilibrium velocity, which is the expected velocity to be approached by the dispersed and continuous phases if the flow field is extended far enough.

$$v_{eq} = \alpha_{c,inlet} v_{c_i nlet} + \alpha_{d,inlet} v_{d,inlet} \quad (26)$$

where a value of 4.96 m/s has been obtained.

The numerical results of the dispersed particles and air velocities are shown in Fig. 9. The air stream velocity can be considered to be nearly constant and the analytical variation of the particles velocity can be obtained again from Eq. (25). Good agreement is obtained between numerical results and analytical solution.

The second case is labeled as dilute bubbly flow. Along the tube two stream are flowing. The continuous phase is considered to be water and the dispersed phase to be air. The difference with previous cases is the value of the densities ($\rho_d = 1.0$ and $\rho_c = 1000$) and the void fraction at the inlet $\alpha_{d,inlet} = 0.1$. The numerical results have been obtained using a time step of 0.00001 s, while the other parameters are maintained equal than in the first case.

The bubble and liquid phase velocities are depicted in Fig. 10a. As can be observed, both phase velocities tend to reach the equilibrium velocity of 4.6 m/s, obtained from Eq. (26), in the first centimetres of the tube length.

A variation of this second case correspond to a dense bubbly flow. In this case, the void fraction value is $\alpha_{d,inlet} = 0.5$. The analytical solution for the equilibrium velocity obtained from Eq. (26) is 3.0 m/s. The evolution of the phase velocities can be seen in Fig. 10b, where the numerical results tend to reach quickly the equilibrium velocity.

7.3. Sedimentation case

This case is a simplified gravity-induced phase separation problem proposed by Coquel et al. (1997). The most important characteristic of the sedimentation problem is the transition from two-phase flow to single-phase flow. The case is also useful to test the stability of the numerical resolution method.

The sedimentation problem consists of a vertical pipe of length 7.5 m where, like the Water Faucet problem, gravity is the only source term considered. Initially, at time $t = 0.0$ s, the pipe is filled with a uniform mixture of liquid and gas, at pressure $p = 10^5$ Pa and a gas void fraction of 0.5. The pipe is closed at both ends. Velocities at the top and the bottom boundaries are considered to be zero. The gravity field provides a separation of the phases for $t > 0$. The solution at $t = \infty$ is composed of a pure gas at rest at the top side, and a liquid at rest at the bottom side of the pipe. A schematic representation of the case is depicted in Fig. 11.

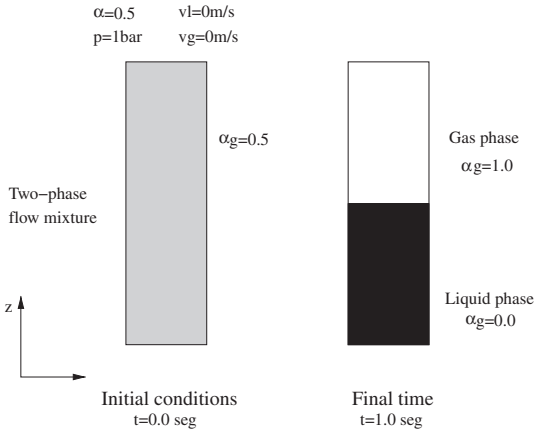


Fig. 11. Sedimentation case.

The results have been obtained using a mesh of 40 CVs, a time step of 0.001 s, an error criteria of 10^{-10}, and assuming the liquid and gas density of 1000 kg/m^3 and 1.0 kg/m^3, respectively.

Numerical results have been compared with an analytical solution obtained from Evje and Flatten (2003). They proposed a set of mathematical expressions for the evaluation of void fraction, liquid velocity and pressure at different instants and positions. The comparison between analytical solution and numerical results

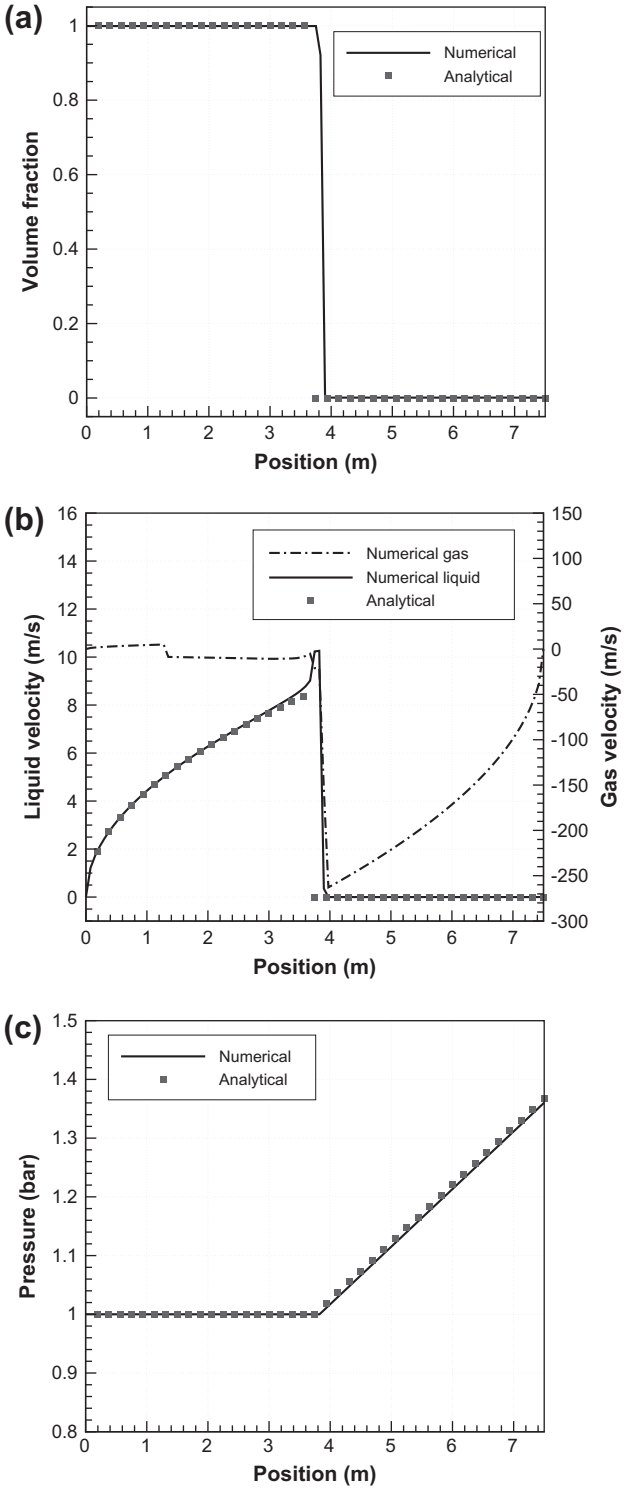


Fig. 12. Sedimentation case: comparison between numerical results and analytical solution at 1.0 s (Evje and Flatten, 2003).

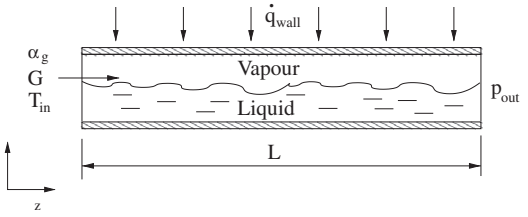


Fig. 13. In-tube evaporator case working with R134a refrigerant.

Table 3
Inlet conditions of the in-tube evaporator.

Working conditions	T_{ev} (°C)	p_{ev} (kPa)	G (kg/m^2 s)	$\dot{q_{ev}}$ (W)
1	−0.86	296	64.78	3573.3
2	−5.50	249	51.80	2985.1
3	−10.13	208	39.96	2418.9
4	−19.73	139	21.95	1143.5

obtained with the model proposed in this paper at time instant 1.0 s are depicted in Fig. 12. Reasonable good agreement can be observed.

7.4. In-tube evaporator case

An in-tube evaporator case has been analyzed by means of the two-fluid model with the aim of checking the influence of considering the energy equation into the model presented. Note that the thermal behaviour has not been considered in the cases presented previously. A schematic representation of the case is depicted in Fig. 13.

An experimental facility, which has been specifically oriented to study vapour refrigerating cycles based on single stage compression units, is used to collect experimental data. The design of the rig was performed to carry out the experimental validation of the numerical modelling of each element which could be found in these systems. Attention is focused on the evaporation process of refrigerant R134a.

The set-up consist of an electrically heated evaporator made of a 6 m length copper tube with an internal diameter of 8.15 mm and a thickness of 0.725 mm. Two Pt-100 platinum resistance sensors measure the fluid flow temperature at the inlet and outlet sections. Ten calibrated K-type thermocouples uniformly spaced along the tube length have been used in the measuring of the refrigerant fluid temperatures. Twenty K-type surface thermocouples equally spaced along the tube wall have been arranged in ten pairs addressed to measuring the wall temperatures at the top and bottom positions of each section. Two 0–6 bars absolute transducers, with an accuracy of ±0.05% F.S., have been fastened to the inlet and outlet cross sections. To measure the pressure drop along the evaporation process, four different pressure taps have been distributed over the evaporation tube. Each tap was connected to a multi-valve manifold array with the purpose of allowing a sequential switch between each one of the taps and the input side of a 0–140 mbar differential pressure transmitter (±0.05% FS accuracy). For details see Raush et al. (2009).

Different inlet conditions have been used to define four working conditions of the in-tube evaporator. Details are shown in Table 3. The flow regime characteristic in all four working conditions is stratified flow. The local heat transfer coefficient and pressure drop are evaluated according to the models described in Section 4.

The numerical results of the in-tube evaporator for the four different working conditions have been used to compare with the experimental data obtained from the facility detailed above. The wall and fluid temperatures, the pressure drop, the void fraction

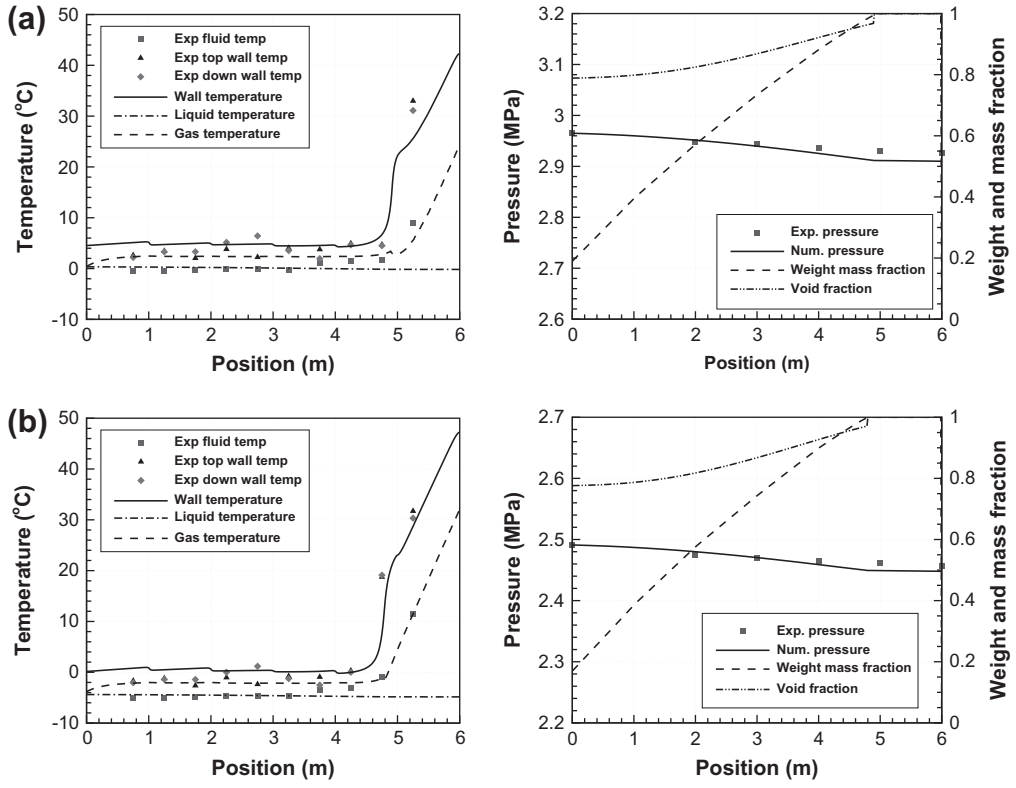


Fig. 14. Numerical and experimental comparison of the temperatures and pressure of working conditions: (a) 1 and (b) 2.

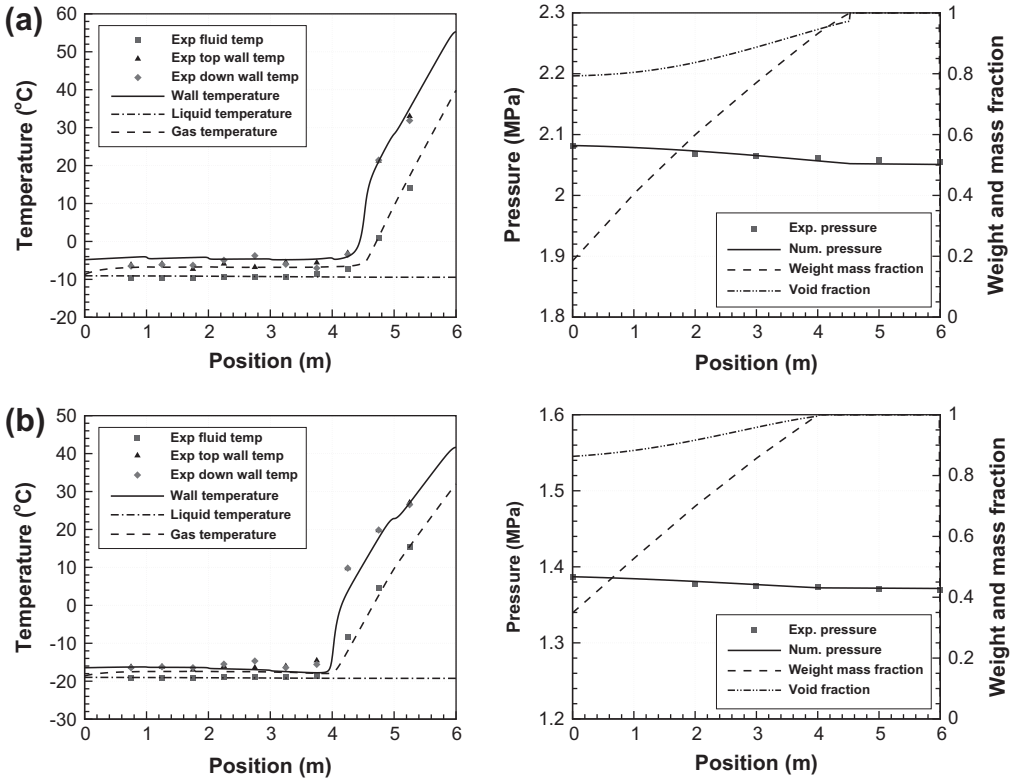


Fig. 15. Numerical and experimental comparison of the temperature and pressure of working conditions: (a) 3 and (b) 4.

and the weight mass fraction profiles of each one of the working conditions are depicted in Figs. 14 and 15. All numerical results have been obtained using a mesh size of 400 CVs, a time step of 0.01 s , and an error criteria of 10^{-6}.

Numerical results over predict wall temperatures in the first 2 m of the in-tube evaporator length for the working conditions 1, 2 and 3. This over-prediction gives a maximum error of 40% for working conditions 1 at 0.8 m of length. However, good agreement between numerical results and experimental data are obtained after 2 m, given a maximum error of 13% for working conditions 1 at 3.8 m of length. The numerical results of the working condition 4 present good agreement along the first 4 m of length, where occurs the two-phase flow phenomena. A maximum error of 2% is found.

The wall temperature difference between numerical results and experimental data is mainly due to the empirical correlation chosen to evaluate the heat transfer coefficient of the phases. To evaluate these parameters two empirical correlations based on stratified flow were used. Although the flow regime map predicts an stratified flow along the in-tube evaporator, there is no security that this regime is presented in the first 2 m of the in-tube evaporator length.

The numerical results of the gas temperature is always near the experimental wall temperature, while the liquid temperature is in a good agreement with the experimental fluid temperature. When the fluid flow has been evaporated completely and a single-phase gas is flowing inside the tube, the numerical results of the fluid flow represents the thermal behaviour described for the experimental data very well.

The numerical pressure drop is in a good agreement with the experimental data. A maximum error value of 6% is obtained with the working condition 1 at a length of 5 m.

Although the agreement between numerical results and experimental data of the temperature and pressure are good in all working conditions, it is important to clarify that the fluid temperature measured is the fluid mixture temperature. It is impossible to obtain data for each one of the phase temperature with the facility described above.

8. Conclusions

This paper has shown a numerical method to solve the two-phase flow inside pipes. The mathematical formulation is based on the two-fluid model, and the governing equations discretization has been carried out by means of the control volume technique. A numerical algorithm capable to solve the couple between velocities and pressure has been implemented, using an extension of the well-known semi-implicit pressure based method SIMPLE. A group of empirical correlations have been implemented into the mathematical model to evaluate the different parameters required in the set of algebraic equations. The instability characteristic of the two-fluid model due to the system of differential equations is non-hyperbolic has been analyzed. After that, validation and verification of the code have been presented with some well-known benchmark cases. The result dependence of using different numerical schemes to evaluate the convective term has been analyzed. No significant difference has been obtained between Upwind, CDS and Smart schemes. The influence of the gravity force in the numerical resolution has been analyzed by means of the Water Faucet and Sedimentation cases, while the influence of the drag force has been studied with the horizontal particles transport cases. All these cases consider adiabatic flow. Therefore, an extra case was proposed to test the thermal effects. An in-tube evaporator working under four different conditions has been used to test the stability of the numerical resolution method when the energy equation is considered. A good agreement between numerical results and the analytical solutions or experimental data depending on the case has been obtained with the model proposed in this paper.

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